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Interactive metal mixture toxicity as an emergent property

Interactive Metal Mixture Toxicity to *Daphnia magna* Populations as an Emergent Property in a

Dynamic Energy Budget Individual-Based Model

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Environmental risk assessment of metal mixtures is challenging due to the large number of possible mixtures and interactions. Mixture toxicity data cannot realistically be generated for all relevant scenarios. Therefore, methods for prediction of mixture toxicity from single-metal toxicity data are needed. We tested how well toxicity of Cu-Ni-Zn mixtures to *Daphnia magna* populations can be predicted with an Individual-based Model based on Dynamic Energy Budget Theory (DEB-IBM), assuming non-interactivity of metals on the physiological level. We exposed *D.*

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Introduction

Ecosystems might be at risk from contamination with mixtures of chemicals. When Environmental Risk Assessment (ERA) takes possible mixture effects into account at all, it relies on the application of relatively simple reference models. A major challenge is the sheer number of mixtures that might be relevant for risk assessment, and the possibility of synergistic and antagonistic effects. Since it is not feasible to test all mixtures of interest experimentally, it is necessary to develop models which allow to extrapolate from single-substance toxicity to mixture toxicity under untested conditions. In addition, environmental protection goals are usually formulated on the level of populations and higher, and therefore such models should allow to account for ecological processes and their interaction with mixture toxicity.

From individuals to populations

For various reasons, toxicity to populations can differ from toxicity to individuals.

Firstly, population-level effects include the combined effects on lethal and sublethal individual-level endpoints. Secondly, in populations, individuals contribute to population-level effects during their entire lifecycle, and thus individual-level toxicity parameters which are specific for exposure duration, such as ECx-values, are of limited use to predict population-level responses. Thirdly, long-term individual-level toxicity tests are typically conducted without food limitation (OECD, 2012). In contrast, individuals in populations share an environment and thus food resources are limited. Limited food availability can result in altered toxicity to individuals (Heugens et al., 2006), but also lead to compensatory dynamics due to toxicant-induced release from intraspecific competition, potentially leading to a lower sensitivity of population-level endpoints (Pereira et al., 2019). In addition to resource competition,

magna populations to Cu, Ni, and Zn and their mixture at a fixed concentration ratio.

We calibrated the DEB-IBM with single-metal data and generated blind predictions of mixture toxicity (population size over time), with account for uncertainty. We compared the predictive performance of the DEB-IBM with respect to mixture effects on population density and population growth rates with that of two reference models applied on the population level, Independent Action (IA) and Concentration Addition (CA). Our inferred physiological Modes of Action (pMoA) differed from literature-reported pMoAs, raising the question whether this is a result of different model selection approaches, intraspecific variability, or whether different pMoAs might actually drive toxicity in a population context. Observed mixture effects were concentration and endpoint dependent. IA was overall more accurate than CA, but CA predicted effects on population growth rate slightly better. The DEB-IBM most accurately predicted effects on 6-week density, including antagonistic effects at high concentrations, which emerged from non-interactivity at the physiological level. Mixture effects on initial population growth rate appear to be more difficult to predict.

To explain why model accuracy is endpoint-dependent, relationships between individual-level and population-level endpoints should be illuminated.

Keywords: Aquatic toxicology, metal toxicity, mixture toxicology, predictive toxicology, toxicity mechanisms, aquatic invertebrates, freshwater toxicology, computational toxicology

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physiological inhibition due to release of semiochemicals, or by physical interaction, can be relevant mechanisms of intraspecific interaction (Gergs et al., 2014).

Synergistic and antagonistic effects

When dealing with mixture effects, it is desirable to identify cases of synergistic or antagonistic effects. Stating that a mixture effect is synergistic or antagonistic is only meaningful if a reference model is defined (Cedergreen, 2014). Two reference models for mixture toxicity are widely established, being Independent Action (IA) and Concentration Addition (CA). Under the IA model, responses y_i , relative to a control, when exposed to a single substance i are combined multiplicatively to generate the predicted response to the mixture y_{IA} :

$$y_{IA} = \prod_{i=1}^n y_i$$

Equation 1

In order for equation 1 to generate meaningful predictions, y_i must be positive. For binary endpoints, this is treating the mixture response as the combined probability of n independent events. For discrete and continuous endpoints the mathematical application is identical.

CA-predicted responses result from the assumption that toxic units TU_{ECx} add up to one in a mixture giving x effect (Loewe & Muischnek, 1926):

$$\sum TU_{ECx} = \sum_{i=1}^n \frac{c_i}{ECx_i} = 1$$

Equation 2

c_i is the exposure concentration of substance i in the mixture and ECx_i is the concentration of i that elicits x percent of the maximum effect when i is tested individually.

Both reference models, by definition, assume non-interactivity of stressors. CA is generally expected to apply when stressors have similar Modes of Action and IA when stressors have dissimilar Modes of Action (Faust et al., 2003). Deviations from reference models might depend on concentration ratios or levels (Jonker et al., 2005; Loureiro et al., 2010). For example, Nys et al. (2015) reported concentration-level dependent synergistic effects of nickel-zinc mixtures on *Daphnia magna* reproduction. Interaction with other ions in the test medium, or lack thereof, can also drive interactive mixture toxicity (Versieren et al., 2014).

Mechanistic effect modelling

With the addition of concentration level- and concentration ratio-dependent deviation factors (Jonker et al., 2005), static descriptions of mixture toxicity can be highly flexible. However, given that statistical models generally do not allow for extrapolation to untested conditions (e.g. different food levels or exposure durations), mechanistic effect modelling (Grimm & Martin, 2013) might be a more promising tool for predictive modelling of mixture toxicity.

To predict toxic effects under untested conditions mechanistically, Dynamic Energy Budget (DEB) Theory (Kooijman, 2010) is a useful framework. The core idea of DEB theory is that, as organisms take up energy, it is assimilated with a given efficiency and allocated to somatic growth, maintenance of structure, maturation and reproduction. A fixed fraction κ of assimilated energy is allocated to somatic growth and maintenance of structure, whereas the remaining fraction $1 - \kappa$ is allocated to reproduction and maturity maintenance. A widely used extension of DEB theory is the implementation of sublethal effects of single metals or mixtures, in terms of effects on a selected physiological Mode of Action (pMoA), which represent a modification of one or more DEB parameters or energy fluxes (Jager et al., 2006). This approach has

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predictions of mixture toxicity, one has to make assumptions on how mixture toxicity might be best represented on the physiological level. In the case of metals, this is further complicated by the fact that they do not have a single, identifiable molecular target. Therefore, making an *a priori* decision on whether metals should be considered as “different target” or “shared target” substances is not straightforward. While detailed knowledge on physiological mechanisms of metal toxicity to *D. magna* is available for some metals (Brix et al., 2017; De Schamphelaere & Janssen, 2002, 2004), this information is not easily converted to appropriate assumptions for mechanistic modelling of mixture effects. Nys et al. (2017) conducted a meta-analysis on the accuracy of IA and CA concerning metal mixture toxicity to three species, and concluded that IA is overall more accurate than CA. However, the involved studies mostly focus on individual-level toxicity, and it is therefore not clear whether the assumption of IA is also valid on the physiological level.

Uncertainty in mechanistic models

Individual-based Models (IBM) facilitate the use of mechanistic effect modelling with respect to ecological interactions (Martin et al., 2012; Pereira et al., 2019; Vlaeminck et al., 2021). Calibration of IBMs typically relies on numerical optimization algorithms, such as Approximate Bayesian Computation (ABC) (van der Vaart et al., 2015). When calibrating a model that includes mixture toxicity, a separate submodel has to be calibrated for each stressor in the mixture. Consequently, when combining the results of those calibrations, their uncertainties are combined in the final model. It is unclear what the consequences of this combined uncertainty are concerning the accuracy and potential bias of IBM-predicted toxic effects. However, using the results of Bayesian optimization algorithms, it is possible to generate model predictions with account for uncertainty. Instead of using a point estimate to generate predictions, one

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been applied to different taxa (Ashauer & Jager, 2018), but particular experience in this context has been gained with the standard test species *Daphnia magna*. Common applications include modelling of time-varying exposure to chemical stressors (Pieters et al., 2006) and analysis of toxicity based on multiple endpoints (Billoir et al., 2007; Jager et al., 2010). Pereira et al. (2019) used an Individual-based model based on DEB theory (DEB-IBM) to extrapolate nickel toxicity from individuals to populations at different temperatures, and correctly predicted the absence of effects on the population level when exposed to the median effective concentration for reproduction on the individual level. (Vlaeminck et al., 2021) applied the same approach to prediction of Copper-Zinc mixture toxicity to *D. magna* populations from individual-level single-metal toxicity.

Jager et al. (2010) proposed an approach to implement mixture toxicity in a DEB context, and demonstrated this using the toxicity of two polycyclic aromatic hydrocarbons to *D. magna*. If stressors have different pMoAs, implementation of mixture toxicity is relatively straightforward, since they can be applied independently (assuming that uptake and elimination processes occur independently). If the stressors have a shared pMoA, mixture toxicity might be implemented based on the assumption of different or shared molecular targets (given that a change in a particular physiological process, such as assimilation efficiency, might occur via different molecular targets (Jager et al., 2006)). We hereafter refer to the “different target” and “shared target” implementations as IA and CA on the physiological level, respectively. In the case study provided by Jager et al. (2010), model selection was done by fitting both models to mixture toxicity data and selecting the model that described the data best. This approach is suitable to identify mechanisms of mixture toxicity that are suggested by the model. However, if the aim is to perform blind

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might repeatedly sample values from the posterior distribution (van der Vaart et al., 2015), evaluate the IBM for each sample and thus map a distribution of parameter values to a distribution of predictions. The application of such an approach has been demonstrated before for toxicokinetic models (Ashauer et al., 2010), but not in combination with individual-based models or mixture toxicity.

Objective

In this study, we aimed to evaluate whether a DEB-IBM approach can be used to predict toxicity of Cu-Ni-Zn mixtures based on the assumption of IA on the physiological level, given constraints in the available data for model calibration, and using only single-metal toxicity data for calibration. We hypothesized that interactive mixture toxicity on the population-level can emerge from non-interactivity on the physiological level. More specifically, we assumed IA on the physiological level because it is the simplest to apply without estimating parameters from mixture data, and based on the existing evidence on individual-level mixture toxicity (Cedergreen, 2014; Nys, Versieren, et al., 2017). We conducted a microcosm population experiment with *D. magna*, exposing populations in a single-ray-design (Cedergreen et al., 2007) to Cu, Ni and Zn at environmentally realistic concentration ratios. We used data from the control and single-metal treatments to estimate DEB and toxicokinetic/toxicodynamic (TKTD) parameters and generated predicted mixture responses based on the assumption of non-interactivity on the physiological level and with account for uncertainty in parameter estimation. We first analyzed observed mixture responses statistically, to establish whether the data provides evidence for interactive mixture effects.

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We then compared DEB-IBM-predicted with observed responses in the mixture treatments, to evaluate the accuracy of blind predictions relative to the accuracy of reference mixture models.

Materials and Methods

Population Experiment

We obtained a clone of *D. magna* from the Laboratory of Aquatic Ecology, Evolution and Conservation, KU Leuven. The clone was originally sampled from Langerodevijver, Belgium (50.828581N, 4.639960E), and was cultured for 3 months in our laboratory before onset of the experiments.

Organisms were cultured in a COMBO (Kilham et al., 1998) medium, modified to represent more environmentally relevant conditions (55 mg CaCl₂·2H₂O L⁻¹, 55 mg MgSO₄·7H₂O L⁻¹, 1 mg H₃BO₃ L⁻¹) and without the addition of N or P. The cultures were kept in a climate controlled room at 20°C and a 16:8 Light-Dark cycle. pH of the medium was adjusted to 7.5 (±0.2) by adding HCl to a final concentration of 0.5 mM.

No further pH buffers were used. Culture media were aerated and allowed to equilibrate for at least 24h before use.

Test media and conditions were identical to the cultures except for the addition of natural DOC (4 mg L⁻¹) to test media instead of EDTA. Natural DOC had been sampled by reverse osmosis in November 2018 from the Schwarzbach stream (East-Belgium, 50.5210522N, 6.205860E), fractionated by acidification, de-ionized via H⁺-cation exchange and freeze-dried (De Schamphelaere et al., 2003). The freeze-dried material was dissolved in 15 mM NaOH and added to the test medium as combined Fe+DOC stock. 10 mL Fe+DOC L⁻¹ were added to reach a FeCl₃·6H₂O concentration of 0.77 mg L⁻¹ in the medium. Reverse osmosis had previously been shown to not affect protectiveness of DOC with respect to Cu and Zn toxicity (De Schamphelaere et

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We exposed populations to a control, Cu, Ni and Zn and their tertiary mixture with five concentrations per exposure type and four replicates per treatment combination. Nominal metal concentrations were arranged in a geometric series with a fixed, environmentally relevant mass ratio of 1:1:3 Cu:Ni:Zn (Van Regenmortel et al., 2017), ranging from 5 to 250 µg Cu and Ni L⁻¹ and 15 to 750 µg Zn L⁻¹.

Speciation and exposure modelling

We converted measured metal concentrations to free ion activities with WHAM VII (Tipping et al., 2011), based on measured pH, DOC, dissolved Na, Mg, K, Ca, Ni, Cu, and Zn and nominal total Cl, SO₄ and CO₃. For specific assumptions, see (Nys, Janssen, et al., 2017). As solute database, we used the updated National Institute of Standard and Technology (NIST) database (Smith et al., 2004). All constants reported in the NIST database were included. As input for model simulations, we used a continuous time-series derived from the WHAM VII-predicted free ion activities. Measured concentrations, however were only available on a weekly basis. To deal with this, we performed a step-interpolation between measurements. The details of these calculations are described in Supporting Information.

The time-series of free-ion activities predicted by WHAM for each treatment are shown in SI Figure 3 and SI Figure 4.

Individual-based Model

All data treatment and modelling except for speciation modelling was done in Julia 1.3.0. A schematic overview of the modelling scheme is given in SI Figure 1.

Our DEB-IBM implementation was derived from Pereira et al. (2019), which had in turn been derived from the generic DEB-IBM implementation by Martin et al. (2012). The toxicity submodel was a combination of the reduced General Unified Threshold Model of Survival (GUTS-RED, Jager et al., 2011) and the DEBtox approach (Jager

et al., 2005). Metal-spiked test media were allowed to equilibrate for at least 48h before use.

We initiated populations with 2 egg-carrying females (age 7 to 21 days) and 8 neonates (<=24h) in 500 mL medium contained in a 1000 mL polypropylene jar (Avamoplast, Lokeren, Belgium). Populations were daily fed a diluted mixture of *Raphidocelis subcapitata* (also known as *Pseudokirchneriella subcapitata*) and *Chlamydomonas reinhardtii* in a 3:1 cell-based ratio, totaling 8×10⁷ cells L⁻¹, corresponding to approximately 2.5 mgC L⁻¹.

Twice per week, we percolated test media over a 200 µm sieve, retaining all *D. magna* individuals. We transferred populations from the sieve to a petri dish with fresh test medium, where we determined population sizes by taking a series of images with a digital camera (EOS200D, Canon, Tokyo, Japan), connected to a laptop computer and controlled by an xml script, and counting animals from the images using the Cell Counter plugin in ImageJ 1.52j. Because we took a series of images from identical positions at identical time-intervals, we could identify immobilized individuals, which were not counted. We then re-combined populations and fresh medium with old medium to reach 25% renewal.

Once per week, we took two integrated 10 mL samples of new and old media per treatment. Both were passed through a 0.45 µm filter (Acrodisc, Pall Life Sciences). One was acidified with HNO₃ to a final concentration of 0.14 M and used for analysis of metal concentrations and major cations (ICP-OES). Details on ICP-OES measurements are given in Supporting Information (SI Table 1-3). The other sample was used for analysis of DOC concentrations (TOC-L, Shimadzu, Kyoto, Japan). At the same occasions, pH was measured using a pH electrode (p407, Consort, Belgium) in old and new media.

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et al., 2006). A major difference between our DEB-IBM implementation and previous implementations was that, *a priori*, we allowed for stressors to act via arbitrary combinations of pMoAs. We decided to do so in order to be able to explain some of the more complex single-metal responses, and because different combinations of pMoAs might have different implications for mixture toxicity on the population-level. In addition to the assumptions that are inherent to DEB, DEBtox and GUTS, we made three additional major assumptions:

Firstly, the model included a single food source. In the population experiment, populations were fed a mixture of two algal species. We did not model those explicitly, but modelled food dynamics as if there was only one type of resource, using total food density (# cells L⁻¹) as input variable. We think this simplification is justified because the composition of the added food mixture was constant throughout the experiment, and because daphnids are, under most circumstances, unselective filter feeders (Lampert, 1974), so it is to be expected that the estimated parameters represent the modelled system equally well across treatments and time-points.

Secondly, we assumed non-interactivity of metals at the physiological level: In reality, metals might mutually affect their uptake and elimination or binding to target sites, and thus interact on a molecular or cellular level. In the model, we assumed metal toxicokinetics and toxicodynamics to be independent. Combined effects on shared pMoAs were applied by assuming IA on the physiological level. Thirdly, we assumed one damage pool per metal: We modelled toxicity with one damage pool per metal, rather than a separate damage pool for every pMoA, thereby avoiding the estimation of a separate k_d value for every pMoA. This means that each metal can have multiple toxicodynamic components, but only one toxicokinetic component.

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Following GUTS-RED, external concentrations $C_{W,i}$ (nM Me²⁺) of metal i were directly translated to scaled damage $D_{W,i}$ (nM Me²⁺) controlled by a single parameter, the dominant rate constant k_d (day⁻¹):

$$\frac{dD_{W,i}}{dt} = k_{d,i} \times (C_{W,i} - D_{W,i})$$

Equation 3

Sublethal and lethal effects were modelled as a function of a common damage pool $D_{W,i}$ for each metal.

In the case of lethal effects, we related $D_{W,i}$ to a hazard rate $h_{z,i}$ (day⁻¹) using a log-logistic equation, and converted $h_{z,i}$ to a stochastic mortality probability $p_{z,i}$.

$$h_{z,i} = \frac{h_{max,i}}{1 + \left(\frac{D_{W,i}}{ED50_{h,i}} \right)^{-\beta_{h,i}}}$$

Equation 4.1

$$p_{z,i} = e^{-h_{z,i}}$$

Equation 4.2

We thus assumed a stochastic death mechanism. In Equation 4.1, we deviate from the original GUTS formulation, where a hockey-stick equation with a threshold concentration and a slope is used to relate damage to hazard rate. The mortality probability $p_{z,i}$ was evaluated independently for each metal. $ED50_{h,i}$ (nM) is the scaled internal damage that leads to 50% of the maximum hazard rate caused by stressor i , $h_{max,i}$; $\beta_{h,i}$ is the corresponding Hill's slope.

For implementation of sublethal effects, we related $D_{W,i}$ to physiological stress S_i specific for pMoA j :

$$S_{i,j} = \frac{S_{max,i,j}}{1 + \left(\frac{D_{W,i}}{ED50_{S,i,j}} \right)^{-\beta_{S,i,j}}}$$

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Calibration

For parameter inference, we used a Sequential Monte Carlo Approximate Bayesian Computation (SMC-ABC) scheme, adapted from previously described algorithms (Sisson et al., 2007; Toni et al., 2009). Details about SMC-ABC are given in supporting information. Prior distributions were initially defined as Log-Normal distributions to conservatively cover the range of biologically plausible values, with standard deviations covering at least one order of magnitude. In some cases, Log-Normal distributions were replaced by narrower Uniform distributions to improve convergence behavior.

DEB parameters were fitted to population density in the control, using values from the Add-my-pet database (Kooijman & Gergs, 2019) as default. TKTD parameters were fitted to $\ln(y + 1)$ -transformed control-normalized densities over time. Because SMC-ABC is particle-based, covariance between parameters was accounted for.

Our model implementation allowed for simultaneous effects via five toxicity components (four pMoAs and direct lethal effects). This means that the SMC-ABC algorithm could converge towards a model where a single pMoA, or any combination of pMoAs drives metal toxicity. After parameter inference, we conducted additional Monte Carlo simulations to evaluate to which extent each component drives toxic effects in the calibrated models. This was done by simulating the single-metal treatments with each component acting separately (detailed scheme given in Supporting Information).

Mixture analysis

We analyzed deviations of observed responses from IA and CA predictions based on two endpoints, initial growth rate r (day⁻¹) and 6-week density (# Daphnids L⁻¹).

Because observed population trajectories did not always fit a logistic growth curve,

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Equation 5

$ED50_{S,i,j}$ is the 50% effective damage of metal i with respect to pMoA j . $S_{max,i,j}$ (→)

is the corresponding maximum stress level. The state variable $S_{i,j}$ modifies

physiological processes based on the physiological Mode of Action (pMoA). We considered the four most commonly considered pMoAs (Jager et al., 2010), being 1), increase in growth costs, 2) increase in somatic and maturity maintenance costs, 3) decrease in assimilation flux, 4) decrease in reproduction efficiency (SI Table 4).

In addition, we fitted a mortality constant m_e which results in increased mortality at reduced scaled reserve density (Martin et al., 2013) and generally dampens oscillatory fluctuations of population density over time. In DEB, the scaled reserve density e (→) is the amount of reserve relative to the maximum amount of reserve, or reserve capacity. It can take values between 0 and 1 and converges to one if an individual's assimilation rate is at its maximum. Based on m_e and e , a mortality probability p_e is calculated at every time-step of the model.

$$p_e = (1 - e)(m_e)^{\frac{1}{TR}}$$

Equation 6

TR (# time-steps day⁻¹) is the temporal resolution at which the IBM is executed.

For further model analysis, we investigated predicted population-averages of some DEB-IBM state variables, namely the scaled ingestion rate $\bar{J}_X$ (# cells day⁻¹), functional response f (→), and assimilation rate $\bar{p}$ (J day⁻¹) (see Supporting Information for more details).

A model description according to the Overview, Design Concepts and Details (ODD) protocol (Grimm et al., 2010) is given in Supporting Information.

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we calculated r based on the peak density N_{peak} , the initial density N_0 and the time to reach N_{peak} : t_{peak} (days).

$$r = \frac{\ln(N_{peak}) - \ln(N_0)}{t_{peak} - t_0}$$

Equation 7

If population density declines monotonically throughout the experiment, the denominator of this equation is 0 and r is not defined. We set r to 0 in these cases, despite the fact that a monotonically decreasing population has in fact a negative growth rate. The reasons for this decision were that firstly, any extinctions that occurred in the population experiment occurred within the first 7 days of exposure, and therefore differentiating between extinctions occurring at different paces (i.e., different negative values of r), was not of particular interest for our analysis. Secondly, negative responses would complicate the generation of IA-predicted responses, since multiplication with negative values is likely to lead to erroneous predictions, while at the same time not providing any additional insight in this particular case.

The IBM-predicted population growth rates were derived from the predicted time-series data the same way that population growth rates were derived from observed time-series data.

For 6-week density, we used the average population density in the last week of the experiment, to make this endpoint less prone to stochastic fluctuations in population density.

IA- and CA-predictions were generated based on relative responses, i.e. population density in a given treatment and at a given time-point divided by the control average at the same time-point. To generate IA-predicted responses y_{IA} , we multiplied

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observed relative responses of the corresponding endpoint to single metals. To generate CA-predicted responses over time, we treated subsets of the data as independent datasets, each subset consisting of the control-normalized population densities in all single-metal treatments at a given time-point. For each metal i at every time-point t in the single-metal tests, we performed nonlinear least squares regression (function *curve_fit* from the LsqFit package in Julia) on the observed relative responses versus WHAM-predicted free-ion activities in the single-metal treatments, to estimate a median effective concentration $EC50_{t,i}$ (nM) and Hill's slope $\beta_{t,i}$ (-) according to the following equation:

$$y_{t,i} = \frac{1}{1 + \left(\frac{C_{t,i}}{EC50_{t,i}} \right)^{\beta_{t,i}}}$$

Equation 8

$C_{t,i}$ is the WHAM-predicted free ion activity of metal i at time-point t . Since the effective concentration ECx is inversely related to the concentration-response function at response x , the CA-predicted response y_{CA} at time-point t was then determined by solving

$$\sum_{i=1}^n \frac{C_{t,i}}{EC50_{t,i} \times \left(\frac{y_{CA,t}}{1 - y_{CA,t}} \right)^{\beta_{t,i}}} = 1$$

Equation 9

numerically (Nys et al., 2017), using SMC-ABC. n is the number of metals in the mixture.

To generate CA-predicted responses of growth rate and 6-week density, we fitted logistic curves to the single-metal responses and solved Equation 9 correspondingly.

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from the posterior distribution in each step. This results in a distribution of predictions for each predicted data point, which maps the uncertainty in parameter estimation to the uncertainty in model output.

To generate predictions from the DEB-IBM, we sampled 100 independent particles from each of the four joint posterior distributions (one for DEB parameters and one for each metal) and simulated the experiment for each sample. The predicted growth rates and 6-week density were calculated for each sample separately.

Results

Physicochemistry

Mean measured physicochemical test media characteristics in the population experiment are reported in Table 1. In the first week of the experiment pH increased up to 9.9, likely due to photosynthetic activity of the added algae, but fell back down when *D. magna* population sizes increased (SI Figure 2).

Across all treatments (single-metals and mixture), mean measured dissolved metal concentrations deviated at most by a factor of 1.8 from nominal concentrations. The average ratio between mean measured and nominal concentrations across all treatments was 0.99 (SD=0.3, N=20). Mean measured metal concentrations in the mixture treatments deviated at most by a factor of 1.9 from measured concentrations in the corresponding single-metal treatments. Metal concentrations in the mixture treatments deviated from the corresponding single-metal treatments by an average factor of 1.08 (SD=0.28, N=15) (see SI Table 5 for all metal concentrations per treatment).

Average Cu concentration in the controls was 5.8 $\mu\text{g L}^{-1}$, which could potentially have contributed somewhat to background toxicity in *D. magna*. However, we have not observed adverse effects of Cu on initial growth rate or 6-week density at measured

To quantify deviations from reference models, we estimated the deviation parameter α for IA and CA (Jonker et al., 2005) with SMC-ABC, by solving equations 9.1 and 9.2 for CA and IA, respectively.

$$\sum_{i=1}^n \frac{C_{t,i}}{ECx_{t,i}} = \exp(G)$$

Equation 10.1

$$y_{IA} = \phi \left(\phi^{-1} \left(\prod_{i=1}^n y_{t,i} \right) + G \right)$$

Equation 10.2

ϕ is the cumulative distribution function of the standard normal distribution and G is the deviation function, which depends on the relative contributions to toxicity

$z_i = \frac{r u_i}{\sum r u_i}$ and the deviation parameter α :

$$G = \alpha \times \prod_{i=1}^n z_i$$

Equation 11

We tested for significant interactions using a nested F-test (Hochmuth et al., 2014). To account for multiple testing, we applied a Bonferroni correction. Given an initial significance level of 0.05 and 20 significance tests conducted, this resulted in a significance level of $\alpha = \frac{0.05}{20} = 0.0025$.

Validation

Since there was only one measured variable available, but a total of 16 parameters to estimate for each metal, we expected our estimation of TKTD parameters to be associated with large epistemic uncertainty. To account for this, we refrained from deriving point estimates of parameters from posterior distributions. Instead, we generated predictions by running Monte Carlo simulations and sampling a particle

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concentrations up to 13.4 Cu $\mu\text{g L}^{-1}$, suggesting that background toxicity of Cu was not an issue for the endpoints of interest. Without individual-level data, it cannot be ruled out entirely that some individual-level background toxicity occurred, which might have been compensated for on the population level. However, given the mean measured media physicochemistry (Table 1), the average bioavailable Cu concentration in the control was only 0.14 $\mu\text{g L}^{-1}$, and the bioavailability-corrected HC5 equal to 38 $\mu\text{g L}^{-1}$ (both calculated in BIO-MET bioavailability tool Version 5-June 2019), which is 7 times above the average Cu concentration in the control.

Calibration

Population dynamics in the control followed a delayed logistic growth pattern, with a peak density of 877 (SD=55, N=4) daphnids L^{-1} at day 18, corresponding to an initial population growth rate of 0.21 day^{-1} . The mean final population density was 447 (SD=15.1, N=4) daphnids L^{-1} . We could approximate this pattern with the DEB-IBM by correcting food-dependent parameters (Table 2, Figure 1). Estimated values for parameters which are also reported in the add-my-pet database were within the range of values reported for different species of the genus *Daphnia*.

The DEB-IBM could be fitted well to single-metal control-normalized population densities. Only at the highest Ni concentration, the relative responses were continuously under-estimated (Figure 2). At intermediate concentration levels, we often observed effects on the initial exponential growth phase of the population, followed by partial or full recovery. A notably complex response pattern was observed at the second-highest Zn treatment (717 nM), where an effect on the exponential growth phase was followed by recovery of the population, which was in turn followed by collapse of the population. This was qualitatively and quantitatively reproduced by

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the DEB-IBM. Predicted responses were also adequate for initial growth rate and 6-week density (Figure 3). Notably, Ni did not affect population growth rates.

Posterior distributions of the estimated TKTD parameters were relatively broad (SI Figure 5-7, SI Table 6-8), but the variability in corresponding predicted responses was acceptable (the normalized root mean square error between fitted and observed values was below 0.1 in all cases), indicating that different parameter combinations predicted similar patterns.

Analysis of pMoAs suggested effects on reproduction efficiency as the main driver of toxicity for all three metals. For Zn, effects on assimilation flux contributed to toxicity at higher concentrations. Direct lethal effects played a subordinate role except at the highest tested concentrations, where they contributed clearly to population extinction due to Cu and Zn (Figure 4). At the highest Zn concentrations, the observed population extinction was predicted to be caused by a combination of all pMoAs.

However, since all populations went extinct within less than a week in these treatments, these subsets of the data do not provide much information on mechanisms of toxicity, and therefore we do not consider the combined effects of pMoAs at high concentrations to be biologically relevant, since direct lethal effects alone might explain this pattern equally well.

Statistical analysis of mixture effects

Significance testing indicated statistically significant ($p < 0.0025$) mixture interactions in all but the highest mixture treatments (Table 3). However, the high statistical significance was in many cases attributable to either a relatively low variability of observed responses, or to the fact that mixture models were fitted to every mixture treatment individually, due to the apparently high concentration-level dependency of deviations. Since the nested F-test is based on the Sum of Squared Errors of the fitted

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Expressed in responses of growth rate and 6-week density, the predictions of mixture effects to 6-week density were substantially more accurate than predictions of effects on growth rate (Table 4). The mixture dose-response of growth rate was not well predicted by any of the models, but slightly better by CA than IA and the DEB-IBM. Notably, the DEB-IBM over-estimated effects on growth rate at low concentrations, where mixture exposure was even observed to be associated with a slight stimulation of growth rates. The synergistic effect on growth rate in the second-highest mixture treatment was only partially predicted. Effects on 6-week density were best predicted by the DEB-IBM, including antagonistic effects at high concentrations (Figure 5). According to the DEB-IBM predictions, exposure to single metals and mixtures was associated with stimulation of average ingestion rates (SI Figure 8-9), with similar effects on assimilation rates under Cu and Ni exposure (SI Figure 10). Under Zn exposure, average assimilation rates were also increased, but to a lesser extent than ingestion rates, due to the direct effect of Zinc on assimilation flux. In the mixture treatments, parts of the stimulating effect on average assimilation rates caused by Cu and Ni appeared to be mitigated by the effect of Zn, suggesting that interactions between pMoAs induce interactive mixture toxicity. More detailed information on predicted individual-level state variables is given in Supporting Information.

Discussion

Calibration

Calibration of a DEB-IBM with population-level data is possible, even if only population densities over time are available, which is traditionally not considered to be ideal to estimate DEB and TKTD parameters (Kooijman et al., 2008; Kooijman & Bedaux, 1996). We acknowledge that population density alone is insufficient to derive parameters that define uptake of food, assimilation and allocation of energy within

model and the null model, there is a high chance that even a small improvement of model fit compared to the null model leads to high statistical significance with this approach. Therefore, high statistical significance does not necessarily indicate toxicological concern.

The clearest interaction was found in the third-highest mixture treatment (Mix3, 29.7 $\mu\text{g Cu L}^{-1}$, 34.2 $\mu\text{g Ni L}^{-1}$, 90.1 $\mu\text{g Zn L}^{-1}$, $\sum TU_{EC50,r} = 0.56$, $\sum TU_{EC50,K} = 1.21$), where effects on growth rate as well as 6-week density were antagonistic. At the second-highest test concentration Mix4 (54.3 $\mu\text{g Cu L}^{-1}$, 68.9 $\mu\text{g Ni L}^{-1}$, 202 $\mu\text{g Zn L}^{-1}$, $\sum TU_{EC50,r} = 1.7$), effects on growth rate were on average synergistic relative to IA and antagonistic relative to CA (Figure 5).

Validation

Overall, the predictive performance of IA, CA and the DEB-IBM was comparable, though relative performance of the three models was highly endpoint- and concentration level-dependent, both in terms of responses of initial growth rate and 6-week density (Figure 5) as well as in terms of responses of population density over time (Figure 6). In the lowest mixture treatment ($\sum TU_{EC50,t} = 0.2$), observed responses over time matched IA- and DEB-IBM-predicted responses. In two mixture treatments ($\sum TU_{EC50,t} = 0.4$, $\sum TU_{EC50,t} = 2.8$), observed responses were lower than IA predictions. This interactive effect was only predicted by the DEB-IBM for the higher treatment. At $\sum TU_{EC50,t} = 1.1$, recovery of the population was observed, which was qualitatively predicted by the DEB-IBM, but delayed in time. Overall, the response over time showed concentration-level dependent deviations from reference models. In the highest mixture treatment ($\sum TU_{EC50,t} = 12.5$), predicting extinction was trivial, since this was already fully explained by single-metal responses to Cu and Zn.

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organisms without uncertainty. To a lesser degree, this is also the case when parameters are estimated from life-table data, as typically generated in the course of ecotoxicological experiments with *Daphnia* (Kooijman et al., 2008). Uncertainty in parameter estimates is to some extent inherent to DEB-based models, since some of the parameters are unmeasurable or only measurable with immense experimental expenditure, and thus can only be estimated with uncertainty. We thus advocate the use of methods that propagate uncertainty when applying DEB-IBMs and similarly abstract models, for example by running Monte Carlo simulations rather than using point estimates of parameters.

Adding to the advantage of the DEB-IBM approach, TKTD parameters are independent of exposure duration. Applying empirical mixture models requires ECx values which do depend on exposure duration (Jager, 2011), so concentration-response curves have to be fitted to every time-point of interest, which is not always a simple task, for example if concentration-response curves are not well described by a simple log-logistic function.

Physiological Modes of Action

Analysis of pMoAs indicated that effects on reproduction efficiency at individual level were a main driver of effects at population level for all three metals. This might be a consequence of the fact that the only available measurement was population density (# daphnids L^{-1}). In the given units, this is most directly reflected by varying reproductive output, rather than growth or maintenance rates. In other words, an effect on reproduction efficiency directly affects the number of individuals in the population, whereas an effect via other pMoAs first affects energy fluxes in other parts of the model, and only indirectly affects the number of individuals in the population, and therefore, the kind of measurement that is available might affect which conclusions

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about pMoAs are drawn. Therefore, the generalizability of the predicted pMoAs can be questioned. However, in the case of Zn, effects on assimilation rate were also essential for predicting the single-metal effects in addition to effects on reproduction efficiency, indicating that physiological processes other than reproduction were also accounted for. Our identified pMoAs were not in agreement with those reported in previous studies on *D. magna*. However, those amount to a relatively small number of studies (Cu and Zn: cf. (Ashauer & Jager, 2018; Vlaeminck et al., 2021), Ni: (Pereira et al., 2019)), and also used different approaches for parameter estimation. A hypothesis that might explain this deviation, and which has not been considered so far, is that pMoAs might be different in a population context than in an individual context, for example due to physiological adaptation of animals under long-term exposure, or mediated by crowding. If this is the case, extrapolation of toxicity across levels of biological organization might be limited with currently advocated approaches, and therefore this hypothesis deserves further investigation.

The approach we used for selection of pMoAs, i.e. to allow for arbitrary combinations of pMoAs, is not common in DEB-based modelling. A disadvantage of this approach is that a larger number of parameters has to be estimated simultaneously. We think that the approach is nevertheless justified from different perspectives: from a practical point of view, an advantage of this approach is that only a single calibration per metal has to be run, and the final posterior distribution encompasses all models that describe the data to some degree. The influence and probability of different pMoAs can be analyzed *a posteriori*, but there is no need to decide for a single pMoA and exclude all other possibilities *a priori*. Furthermore, effects of Zn were in our case clearly best described by a combination of two pMoAs. At the same time, we have seen that if a single pMoA sufficiently explains the data, the SMC-ABC algorithm will converge

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increase, on the other hand, is sensitive to initial conditions. Consequently, predictions of the rate of increase can more easily suffer from inaccuracies in the definition of initial conditions in the DEB-IBM. More precisely, our initial populations contained adult females, and thus the life-history of those individuals, prior to the onset of the experiment, has to be taken into account in the simulations. While we attempted to do so by predicting individual state variables of adult females under culture conditions, we did not have information on some crucial aspects such as the average number of eggs in the brood pouch at day zero. If growth rates are of interest, we therefore suggest to initiate populations with juveniles only, possibly of varying age to minimize fluctuations in population density due to synchronized reproduction. Alternatively, additional data about the life-history of initial adults may be recorded, in order to ensure that those are translated to the corresponding state variables of initial adults in the simulation.

Secondly, the calculation of rate of increase is always influenced by data-points with low population densities. These are subject to higher stochasticity and therefore, growth rates can be expected to be associated with higher variability than 6-week density and thus be harder to predict.

Taking these factors into account, however, it is still unclear why the DEB-IBM over-estimated effects on growth rate at low concentrations, and was unable to predict synergistic effects at high concentrations. We think that relationships between individual-level parameters and (mixture) effects on growth rate first have to be further clarified to explain this trend.

Analysis of predicted ingestion rates, assimilation rates and fecundity indicated that overall, compensation effects due to mitigation of food competition played an important role for (mixture) toxicity on the population-level. This also explains why

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toward that model, as was the case for Cu and Ni. I.e., the model will not necessarily be over-fitted, as one might expect if the potential model complexity is high. In addition, a more detailed model analysis (SI Figure 8, SI Figure 9) suggested that the interplay between simultaneously acting pMoAs was important for the prediction of interactive mixture effects.

It has to be noted that, given the ratio between estimated parameters and data-points in this calibration exercise, there is considerable uncertainty around the TKTD parameter estimates, as well as around the identification of pMoAs, and therefore the application of such a model should go hand-in-hand with uncertainty propagation techniques. As stated before, this uncertainty should caution against generalization of the pMoAs identified in this study to other model applications. We therefore present this modelling approach as proof of principle, showing the capacity of the DEB-IBM approach to predict mixture effects under limited data availability when combined with uncertainty propagation.

Predicting mixture effects to population-level endpoints

Based on the assumption of non-interactivity of metals on the physiological level, we could successfully apply the DEB-IBM approach to perform blind predictions of metal mixture toxicity to population-level endpoints, including correct predictions of antagonistic effects that resulted from the assumption of non-interactivity on the physiological level.

Predictions of mixture effects on 6-week density were substantially more accurate and more precise than those of effects on population growth rate. This discrepancy might be attributed to two inherent differences between these endpoints:

Firstly, 6-week density is an ergodic property of the system, i.e. does not change as a function of initial conditions when food and toxicant levels are kept constant. Rate of

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for growth rate and 6-week density, different patterns of mixture toxicity can emerge:

By the time that the population reaches steady state, food is always limiting, and therefore food limitation has a large influence on effects on 6-week density. In contrast, the initial growth rate encapsulates the time-window of the experiment where food is not limiting, or only starts to become limiting. Consequently, interactions between food competition and effects on growth rate vs effects on 6-week density should also be expected to be different.

Assuming non-interactivity of metals

The DEB-IBM predicted antagonistic effects on 6-week density relative to IA, despite IA being assumed on the physiological level. This predicted antagonism was in line with observed effects, which shows that interactive effects do not have to be implemented on the physiological level in order to explain interactive mixture effects on higher levels of organization. Analysis of individual-level state variables suggested that the predicted antagonistic effect was a result of the combined effects of metals via different pMoAs, namely reproduction efficiency and assimilation rate.

Our results indicate that the assumption of non-interactivity on the physiological level can be used in order to predict population-level metal mixture effects. However, a synergistic effect on growth rate was only partially predicted. Failure of the DEB-IBM to predict synergistic effects could point at interactive mixture toxicity on the individual level, which might be more important with respect to effects to population growth rate, compared to 6-week population density, given that population growth rate is in general less strongly influenced by density dependent processes, and therefore it is less likely that an interactive effect will emerge from non-interactivity on the individual level.

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Conclusions

We showed that metal mixture toxicity to *D. magna* populations is in large part predictable based on the assumption of non-interactivity on the physiological level. While further validation is needed before generalizing this conclusion, our findings are in agreement with previous analyses of metal mixture toxicity, showing that strong interactive effects between metals are rare (Cedergreen, 2014) and that IA is, on average, the more adequate reference model than CA at the individual level (Nys, Versieren, et al., 2017). If confirmed by further validation, the use of DEB-IBM in mixture ecotoxicology opens up the opportunity for extrapolations that are not possible with statistical reference models, such as prediction of mixture toxicity at different food levels or exposure durations. Therefore, mechanistic effect modelling can provide considerable advantages over purely empirical models even if the predictive performance of the mechanistic model does not exceed that of empirical models. Furthermore, mechanistic models can serve as a tool for the formulation of clear and well-founded hypotheses, since they point at specific processes being responsible for observed phenomena.

Accurate predictions can be provided without using point estimates of TKTD parameters. Instead, by sampling parameter values from joint posterior distributions, the uncertainty associated with each estimation can be propagated. Also in terms of pMoAs, similarly probable alternative models can be taken into account when generating predictions. This facilitates the use of mechanistic effect modelling under uncertainty, e.g. when not all of the desired data for calibration are available. However, it should raise caution when trying to make generalized statements about pMoAs.

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To explain why DEB-IBM predictions are much more accurate for 6-week density than for growth rate, we suggest that relationships between individual-level processes and population-level endpoints should be further illuminated. Specifically, our results suggest that implementation of interactive mixture toxicity on the individual-level might be more important when the goal is to predict mixture effects on population growth rates.

Our identified pMoAs were not in agreement with those reported in previous studies. For the advancement of mechanistic effect modelling approaches, it will be important to understand whether this is the result of intraspecific variability, an artifact of different model selection approaches, or whether pMoAs might actually shift under toxicant exposure in a population context.

Supplemental Data—The Supplemental Data are available on the Wiley Online Library at DOI: 10.1002/etc.xxxx.

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Data availability statement—Data and code are available via Figshare. Code and usage instructions are additionally available via the GitHub profile of the lead author. Data, associated metadata, and calculation tools are available from the corresponding author (simon.hansul@ugent.be).

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FIGURE CAPTIONS

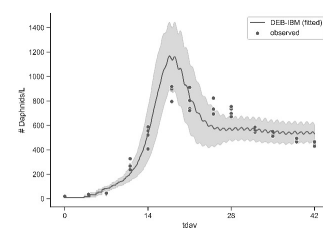


Figure 1 Fitted and observed population dynamics in the control. Lines and ribbons show mean and standard deviation of predictions generated from 100 posterior samples.

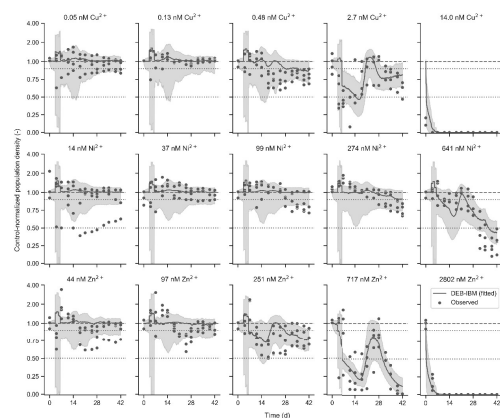


Figure 2 Fitted and observed control-normalized responses of population density in single-metal treatments. Responses above one (dashed gray line) are shown on a logarithmic scale. Fitted values are the result of evaluating the DEB-IBM for 100 samples from the corresponding posterior distribution of TKTD parameters, as well as the posterior distribution of DEB parameters. Lines and shaded areas indicate the mean and standard deviation of predicted responses.

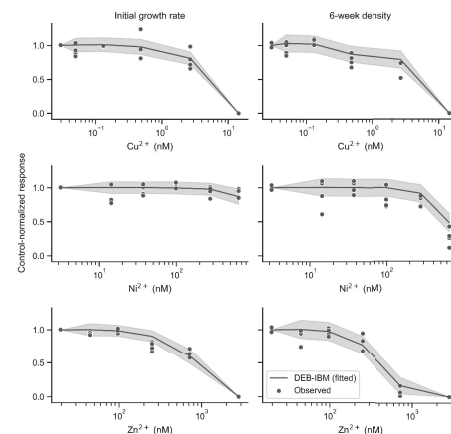


Figure 3 Observed and IBM-predicted control-normalized responses of initial growth rate and 6-week density in the single-metal treatments, after fitting the DEB-IBM to time series of control-normalized population densities in the single-metal treatments. Lines and shaded areas indicate mean and standard deviation of predicted responses based on 100 posterior samples.

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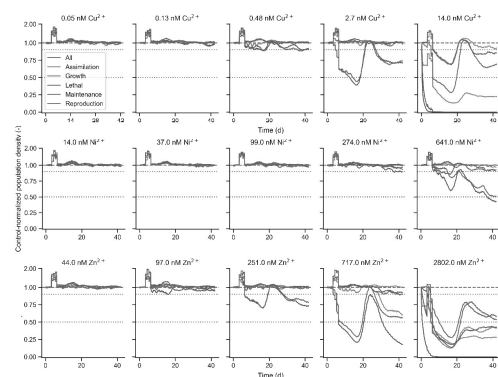


Figure 4 Predicted control-normalized response of population density over time, if all or only one toxicity component is simulated. In the model, effects on reproduction efficiency caused most of the effect across metals at low and intermediate effect concentrations. Responses above one (dashed line) are shown on a log-scale. Dotted lines indicate the 10% and 50% effect levels, respectively.

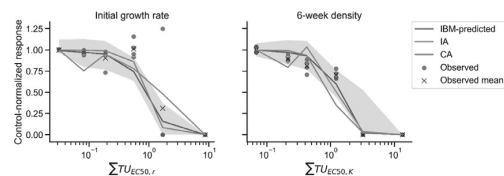


Figure 5 Observed and model-predicted control-normalized responses of initial growth rate and 6-week density in the mixture treatments. Toxic units are calculated based on the corresponding EC50s of individual metals, which were obtained by

fitting log-logistic curves. “IA” and “CA” are empirical models applied on the population-level. “IBM-predicted” are DEB-IBM predicted values based on calibration with single-metal toxicity data and IA on the physiological level.

Table 1 Mean and standard deviation with sample sizes of measured test media physicochemistry

T (°C)	pH	Na _a	Mg _a	K _a	Ca _a	DOC _a
19.4	7.6	27.1	5.01	4.26	13.8	7.32
(0.9,	(0.6,	(2.5,	(0.24,	(1.70,	(0.8,	(0.84,
N=12)	N=268)	N=404)	N=404)	N=404)	N=404)	N=404)

a mg L⁻¹

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Table 2 Estimated DEB parameters for *D. magna*. Point estimates are the particle with the best fit. Credible Limits (CL) indicate the corresponding percentile of the marginal posterior distribution

PARAMETER	UNIT	DESCRIPTION	Add-my-pet reported value (Kooijman & Gergs, 2019)	ESTIMATE	CL ₅	CL ₉₅
p_{Am}	$\frac{J}{cm^2 \times day}$	maximum specific assimilation rate		313.2	260.4	264.0 308.1
F_m	$\frac{L}{cm^2 \times day}$	filtration rate		30.17	10.0	8.6 11.8
$\kappa_{E,X}$	$\frac{J}{algal\ cell}$	assimilation efficiency	not reported	3.7×10^{-6}	3.1×10^{-6}	4.1×10^{-6}
m_e	-	reserve density-related mortality	not reported	0.41	0.32	0.62

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		1	1	1					
	1.70	CA			0.31				<0.002
r			1.34	1.32	1.36	(0.62)	0.08	5	A
	3.21								<0.002
K			-81.2	-87.7	-74.39	0.0 (0.0)	0.02	5	S
Mix5									
r	8.6	IA	Na _{Nr}	Na _{Nr}	Na _{Nr}	0.0 (0.0)	0	-	-
K	13.3		Na _{Nr}	Na _{Nr}	Na _{Nr}	0.0 (0.0)	0	-	-
r	8.6	CA	Na _{Nr}	Na _{Nr}	Na _{Nr}	0.0 (0.0)	0	-	-
K	13.3		Na _{Nr}	Na _{Nr}	Na _{Nr}	0.0 (0.0)	0	-	-

^a Sum of toxic units based on median effect concentration with respect to the corresponding endpoint.

^b Percentile of marginal posterior distribution of a .

^c Mean and standard deviation of control-normalized responses; N=4 for each entry

^d Derived by nested F-test.

^e S=Synergism, A=Antagonism

^f NaN = "Not a Number". Deviation parameter could not be determined if all observed and predicted responses are equal to 0.

Table 4 Model performance per endpoint. SSE is the sum of squared errors between predicted and observed control-normalized values.

endpoint	model	R ² _a	SSE _b
r	IA _c	0.70	1.79
	CA _c	0.75	1.58
	IBM	0.73	1.64
K	IA _c	0.92	0.44
	CA _c	0.90	0.65
	IBM	0.97	0.17

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Table 3 Deviation parameter a (Equation 11) per treatment, endpoint and reference model.

treatment	endpoint	model	$\sum TU_{EC50}$			predict ed by			S/A
			a	CL ₅	CL ₉₅	observe d	null model	p-value _d	
Mix1	r	IA	0.08	6.69	6.63	0.98 (0.03)	0.75	<0.002	A
			0.21	31.6	31.3	0.89 (0.03)	0.8	<0.002	A
	K	CA	0.08	-5.3	-5.36	0.98 (0.03)	1.0	<0.002	S
			0.21	-65.9	-66.6	0.89 (0.03)	0.99	<0.002	S
	r	IA	0.19	-1.35	-1.36	0.9 (0.12)	0.95	<0.002	S
			0.42	-108	-110	0.81 (0.08)	1	<0.002	S
Mix2	K	CA	0.19	-3.81	-3.85	0.9 (0.12)	0.99	<0.002	S
			0.42	-19.6	-19.8	0.81 (0.08)	0.94	<0.002	S
	r	IA	0.56	40.3	34.5	1.01 (0.11)	0.78	<0.002	A
			1.21	25.2	24.9	0.71 (0.05)	0.51	<0.002	A
	K	CA	0.56	17	11.5	1.01 (0.11)	0.86	<0.002	A
			1.21	22.9	22.7	0.71 (0.05)	0.35	<0.002	A
Mix3	r	IA	1.70	-1.77	-1.79	0.31 (0.62)	0.48	<0.002	S
			3.21	-	-	-	-	<0.002	S
	K			154.5	154.5	0.0 (0.0)	0.04	<0.002	S

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^a Linear regression between predicted and observed control-normalized responses in the mixture treatments.

^b Sum of Squared Errors between predicted and observed control-normalized responses.

^c Null model (no deviation parameter)

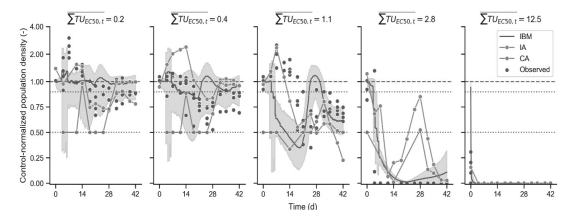


Figure 6 Observed and model-predicted control-normalized responses of initial growth rate and 6-week density in the mixture treatments. Responses above one (dashed gray line) are shown on a logarithmic scale. Toxic units are calculated based on the corresponding EC50s of individual metals, which were obtained by fitting logistic curves. "IA" and "CA" are empirical models applied on the population-level. "IBM-predicted" are DEB-IBM predicted values based on calibration with single-metal toxicity data and IA on the physiological level. Dotted gray lines indicate the 10% and 50% effect levels, respectively.

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